

Computer Vision - 2026

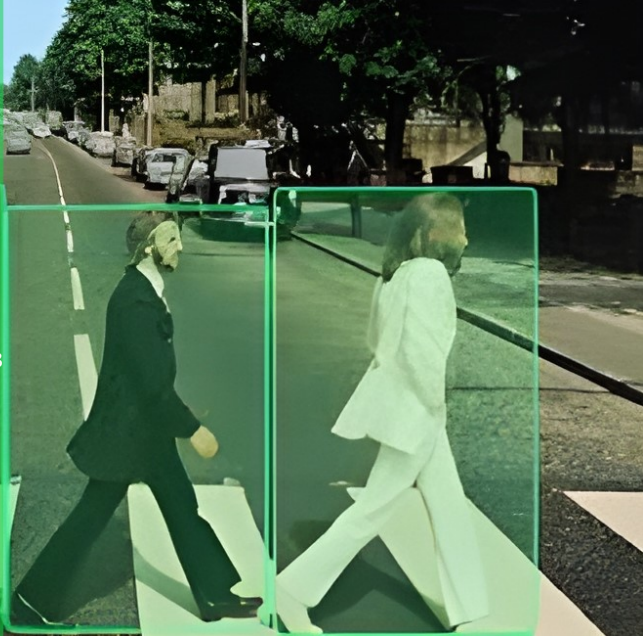
Week #05. Vision-Language Models

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Agenda

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Outcomes

VLMs
Changed CV

SSL Prelude:
From SimCLR
to CLIP

CLIP Deep
Dive

VLM's
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Fine-tuning of
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Three DNNs Training Paradigms

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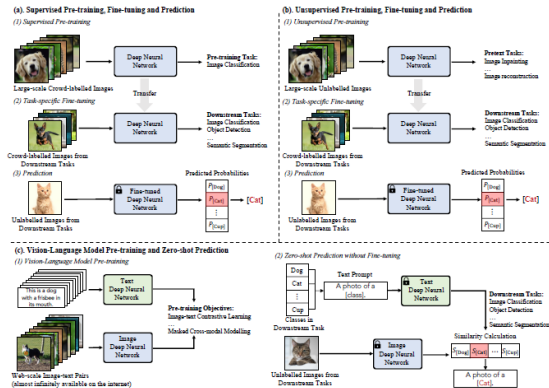


Figure: Three DNN training paradigms in visual recognition. Compared with the paradigms in (a) and (b) that requires fine-tuning for each specific task with task-specific labelled data, the new learning paradigm with VLMs in (c) enables effective usage of web data and zero-shot predictions without task-specific fine-tuning [Zhang et al., 2024]

Pre-Training → Transfer → Distillation

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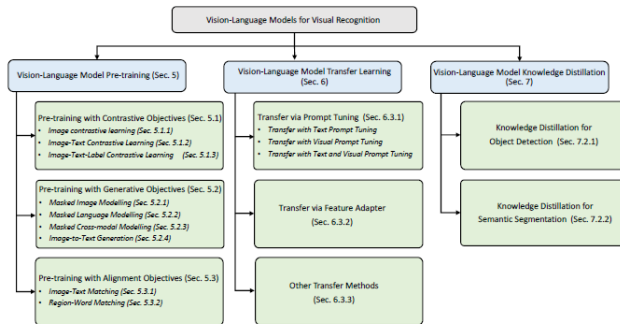


Figure: Typology of vision-language models for visual recognition [Zhang et al., 2024]

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Section 1. Outcomes

Outcomes

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By the end of the week students will be able to:

- 1 **Explain why VLMs changed CV:** open-vocabulary recognition, retrieval, grounding, and “prompting as an API”. InfoNCE/CLIP symmetric contrastive loss and conditional language modeling loss for MLLMs.
- 2 **Run CLIP in practice:** zero-shot classification and image↔text retrieval via prompt engineering and prompt ensembling.
- 3 **Adapt a VLM with PEFT (LoRA):** choose what to tune (projector / vision encoder / LLM blocks) for a small domain dataset.
- 4 **Evaluate responsibly:** report appropriate metrics and identify common traps (prompt sensitivity, leakage/contamination, overclaiming, hallucination).

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Section 2. VLMs Changed CV

Why VLMs Changed CV

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Thesis

Vision-Language Models (VLMs) turned **language into an interface** for vision:

- **Open vocabulary:** concepts are described by text, not fixed class IDs.
- **Transfer at scale:** one pretrained model → many downstream tasks.
- **Prompting as an API:** new tasks can be expressed with prompts/templates.

Key consequence

CV shifted from **task-specific supervised training** to **foundation pretraining + lightweight adaptation**.

From Closed-Set Labels to Open Vocabulary

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Classical CV (Closed-set)

- Fixed label set: $y \in \{1, \dots, K\}$
- Train for one task / dataset
- Generalization limited by K and labeling

$$\hat{y} = \arg \max_k p_{\theta}(y = k | x)$$

Interpretation

Labels become language. The model matches images to textual descriptions rather than predicting a fixed ID.

VLMs (Open-vocabulary)

- Concepts are **text strings** t_k
- Choose labels at inference time
- Zero-shot / few-shot transfer via prompts

$$\hat{k} = \arg \max_k s(f_I(x), f_T(t_k))$$

What VLMs Enable

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Core capabilities

- **Retrieval:** image $\leftrightarrow$ text search in a shared embedding space
- **Zero-shot classification:** prompted text labels act as classifier weights
- **Grounding:** connect words/phrases to image regions
- **Generation & QA:** answer questions about images; follow instructions

Practical view

A pretrained VLM is a reusable **visual feature** + **language interface**:

Image $\rightarrow$ representation $\rightarrow$ (prompted) task output

Two Dominant Model Families

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(1) Contrastive Dual-Encoder

CLIP-like

- Two encoders: $f_I(x)$, $f_T(t)$
- Learn alignment via contrastive loss
- Fast retrieval / zero-shot classif.

$$s_{ij} = \frac{f_I(x_i)^\top f_T(t_j)}{\|f_I(x_i)\| \|f_T(t_j)\|}$$

(2) Generative Multimodal LLM

LLaVA/BLIP-2/Qwen-VL style

- Vis. encoder $\rightarrow$ projector/bridge $\rightarrow$ LLM
- Trained for instruction following / generation
- Better for QA, reasoning-like outputs

$$p_\theta(y_{1:L} \mid x, \text{prompt})$$

Prompting as an API

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Prompted classification with a dual-encoder

Given candidate classes described by prompts $\{t_k\}_{k=1}^K$:

$$\hat{k} = \arg \max_k \frac{\exp(s(f_I(x), f_T(t_k))/\tau)}{\sum_{j=1}^K \exp(s(f_I(x), f_T(t_j))/\tau)}$$

Prompt engineering knobs (low-cost, high impact)

- Template: "a photo of a {class}", "an image of {class}" → prompt ensembling
- Synonyms: multiple names per class → robustness
- Context: domain hints (*medical, satellite, industrial*) → better alignment

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Section 3. SSL Prelude: From SimCLR to CLIP

Why Self-Supervised Learning (SSL)?

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Problem

Large-scale labeled datasets are expensive and narrow. We need representations that generalize beyond fixed label sets.

SSL principle

Create a **pretext task** where supervision comes from the data itself.

- Predict masked content (autoencoding, MLM, MIM)
- Match different views of the same instance (contrastive learning)

Core idea

Learn invariances first, attach task-specific heads later.

Contrastive Learning Intuition

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Instance discrimination

Each sample x_i is treated as its own class.

Pull together: $(x_i^{(1)}, x_i^{(2)})$ Push apart: $(x_i, x_j), i \neq j$

Geometric view

- Positive pairs: same underlying object, different augmentations
- Negatives: different objects
- Goal: structured embedding space

SimCLR Pipeline

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Architecture **Chen et al. [2020]**

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$$x \xrightarrow{\text{augment}} x^{(1)}, x^{(2)} \rightarrow f_{\theta}(\cdot) \rightarrow g_{\phi}(\cdot) \rightarrow z$$

- f_{θ} : backbone encoder (e.g., ResNet)
- g_{ϕ} : projection head (MLP)
- z : normalized embedding

Important detail

The projection head g_{ϕ} improves contrastive learning but is discarded after pretraining.

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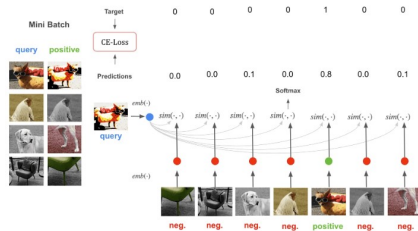
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Similarity: $s_{ij} = \frac{z_i^\top z_j}{\|z_i\| \|z_j\|}$;

Loss for one positive pair (i, j) :

$$\mathcal{L}_i = -\log \frac{\exp(s_{ij}/\tau)}{\sum_{k \neq i} \exp(s_{ik}/\tau)};$$

Total loss: $\mathcal{L} = \frac{1}{2N} \sum_{i=1}^{2N} \mathcal{L}_i$.



Positive pair: two augmented views of the same image.
Negatives: all other samples in the batch [Artemyev and Ashukha \[2024\]](#).

Why Batch Size Matters

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Observation

All other samples in the batch act as negatives.

$$\text{Negatives per sample} = 2N - 2$$

- Larger batch $\Rightarrow$ more negatives
- Better approximation of full contrastive distribution
- Stronger separation in embedding space

Scaling insight

CLIP later scales this idea to tens of thousands of in-batch negatives.

Temperature Parameter τ

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$$p_{ij} = \frac{\exp(s_{ij}/\tau)}{\sum_k \exp(s_{ik}/\tau)}$$

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- Small $\tau \Rightarrow$ sharp distribution (focus on hardest negatives)
- Large $\tau \Rightarrow$ smoother distribution

Interpretation

τ controls the geometry of the embedding space:

- Too small $\Rightarrow$ unstable gradients
- Too large $\Rightarrow$ weak separation

Training Algorithm (SimCLR) **Chen et al. [2020]**

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Algorithm 1 SimCLR Training

```
1: for each minibatch  $\{x_i\}_{i=1}^N$  do
2:   Generate two stochastic augmentations per image:  $\tilde{x}_i^{(1)}, \tilde{x}_i^{(2)}$ 
3:   Encode all views:  $h = f_\theta(\tilde{x})$ 
4:   Project representations:  $z = g_\phi(h)$ 
5:   Normalize embeddings:  $z \leftarrow \frac{z}{\|z\|_2}$ 
6:   Compute similarity matrix:  $s_{ij} = z_i^\top z_j$ 
7:   Compute InfoNCE loss
8:   Update parameters  $\theta, \phi$  using SGD/Adam
9: end for
```

After pretraining, discard the projection head g_ϕ and retain the encoder f_θ for downstream tasks.

From SimCLR to CLIP

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Key observation

SimCLR aligns two **views of the same image**.

Replace view 2 with text.

CLIP idea

$$x_i \rightarrow f_I(x_i) \quad t_i \rightarrow f_T(t_i)$$

$\mathcal{L}_{\text{CLIP}}$ = contrastive alignment between image and text embeddings

Conceptual bridge

CLIP = cross-modal SimCLR at web scale.

What Changed from SimCLR (Image-Image) to CLIP (ImageText)?

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Aspect	SimCLR	CLIP
Modalities aligned	Image - Image	Image - Text
Positive pairs	Augmented views of the same image	Naturally co-occurring image-text pairs
Negative pairs	Other images in the same batch	All other images and texts in the same batch
Training scale	ImageNet-scale (1M images)	Web-scale (400M image-text pairs)
Primary goal	Learn invariant visual representations	Align vision and language in a shared embedding space

Important

The mathematical core (InfoNCE loss) remains the same. The shift in modality pairing and massive data scale enables cross-modal retrieval and zero-shot transfer.

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Section 4. CLIP Deep Dive

CLIP: Core Idea

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Objective

Learn a shared embedding space for images and text using natural language supervision.

$$x_i \rightarrow f_I(x_i) \in \mathbb{R}^d \quad t_i \rightarrow f_T(t_i) \in \mathbb{R}^d$$

$$s_{ij} = \frac{f_I(x_i)^\top f_T(t_j)}{\|f_I(x_i)\| \|f_T(t_j)\|}$$

Key Property

Images and captions describing the same content have high cosine similarity.

CLIP Architecture

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Dual Encoder Design

- Image encoder (ResNet / ViT)
- Text encoder (Transformer)
- Projection to shared normalized embedding space
- Learned temperature parameter τ

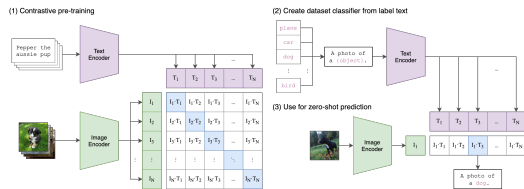


Figure: Contrastive Language-Image Pre-training [Radford et al., 2021]

CLIP Loss: Symmetric Cross-Entropy for Contrastive Learning

Core Idea

CLIP loss is the average of two symmetric cross-entropy (CE) objectives:

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2} \left(\mathcal{L}_{\text{CE}}^{I \rightarrow T} + \mathcal{L}_{\text{CE}}^{T \rightarrow I} \right).$$

Mathematical Formulation

For a batch of N image-text pairs $\{(I_i, T_i)\}_{i=1}^N$, with similarity $s(\cdot, \cdot) = \text{cosine}(\cdot, \cdot)$ and temperature $\tau > 0$:

- **Image-to-Text CE:**

$$\mathcal{L}_{\text{CE}}^{I \rightarrow T} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s(I_i, T_i)/\tau)}{\sum_{j=1}^N \exp(s(I_i, T_j)/\tau)}.$$

- **Text-to-Image CE:**

$$\mathcal{L}_{\text{CE}}^{T \rightarrow I} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s(T_i, I_i)/\tau)}{\sum_{j=1}^N \exp(s(T_j, I_i)/\tau)}.$$

Why Sum Both Directions?

- Prevents modality collapse (e.g., one encoder dominating the shared space).
- Ensures alignment is bidirectional—critical for zero-shot retrieval in both image-text and text-image directions.

Paper reading

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Learning Transferable Visual Models From Natural Language Supervision

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study the performance of this approach by benchmarking on over 30 different existing computer vision datasets, spanning tasks such as OCR, action recognition in videos, geo-localization, and many types of fine-grained object classification. The model transfers non-trivially to most tasks and is often competitive with a fully supervised baseline without the need for any dataset specific training. For instance, we match the accuracy of the original ResNet-50 on ImageNet zero-shot without needing to use any of the 1.28 million training examples it was trained on. We release our code and pre-trained model weights at this [https URL](https://github.com/openai/CLIP). [Radford et al., 2021].

CLIP Loss: an example

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Example: Batch of 2 Image-Text Pairs

Pairs: (Dog Image, "Dog"), (Cat Image, "Cat")

Similarity Matrix:

$$S = \begin{bmatrix} 1.0 & 0.2 \\ 0.1 & 0.9 \end{bmatrix} \quad (\tau = 0.07)$$

Image $\rightarrow$ Text Loss for Dog Image:

$$-\log \frac{e^{1.0/0.07}}{e^{1.0/0.07} + e^{0.2/0.07}} \approx -\log \frac{e^{14.28}}{e^{14.28} + e^{2.85}} \approx 0$$

Text $\rightarrow$ Image Loss for "Cat":

$$-\log \frac{e^{0.9/0.07}}{e^{0.1/0.07} + e^{0.9/0.07}} \approx -\log \frac{e^{12.85}}{e^{1.42} + e^{12.85}} \approx 0$$

Practical Insights for CLIP Loss

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Hyperparameter Sensitivity

- **Batch Size:** Larger N -> better performance (more negatives). CLIP used $N = 32,768$
- **Temperature (τ):** Controls "peakiness" of softmax. Too high -> underfitting; too low -> overconfidence.

Example: Impact of τ

For $s(I, T) = 1.0$ and $s(I, T_{\text{neg}}) = 0.2$:

$$\tau = 0.07 : \frac{e^{14.28}}{e^{14.28} + e^{2.85}} \approx 0.999$$

$$\tau = 1.0 : \frac{e^{1.0}}{e^{1.0} + e^{0.2}} \approx 0.67$$

Smaller τ amplifies differences between positive/negative.

CLIP Loss Example: Small Batch (N=2)

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Setup

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Batch size $N = 2$, temperature $\tau = 0.07$; **image embeddings**: $l_1 = [0.8, 0.6]$, $l_2 = [0.6, 0.8]$; **text embeddings**: $T_1 = [0.7, 0.7]$, $T_2 = [0.7, -0.7]$; normalized embeddings: $\|l\| = \|T\| = 1$; **similarity Matrix**:

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$$S = \begin{bmatrix} l_1 \cdot T_1 & l_1 \cdot T_2 \\ l_2 \cdot T_1 & l_2 \cdot T_2 \end{bmatrix} = \begin{bmatrix} 0.98 & 0.14 \\ 0.98 & -0.14 \end{bmatrix}.$$

Step 1: Image $\rightarrow$ Text Loss (for l_1)

$$\text{Softmax}_{\tau}(S_{l_1}) = \frac{e^{0.98/0.07}}{e^{0.98/0.07} + e^{0.14/0.07}} = \frac{e^{14}}{e^{14} + e^2} \approx 1.0,$$
$$\mathcal{L}_{CE}(l_1) = -\log(1.0) \approx 0.$$

Step 2: Text $\rightarrow$ Image Loss (for T_2)

$$\text{Softmax}_{\tau}(S_{T_2}) = \frac{e^{-0.14/0.07}}{e^{0.14/0.07} + e^{-0.14/0.07}} = \frac{e^{-2}}{e^2 + e^{-2}} \approx 0.018,$$
$$\mathcal{L}_{CE}(T_2) = -\log(0.018) \approx 4.0.$$

Total Loss

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2 \times 2} (0 + 4.0 + \dots) \quad (\text{Sum over all pairs}).$$

Temperature Scaling (τ) Demonstration

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Same Similarities, Different τ

$$S = \begin{bmatrix} 0.98 & 0.14 \\ 0.98 & -0.14 \end{bmatrix}$$

$\tau = 0.07$ (CLIP Default)

$$\text{Softmax}(S_{l_1}) = [0.999, 0.001]$$

Sharp distribution: Focuses on hardest negatives.

$\tau = 1.0$

$$\text{Softmax}(S_{l_1}) = [0.67, 0.33]$$

Softer distribution: Treats negatives more equally.

Implications

- Low τ : Good for clean data, high confidence pairs.
- High τ : Robust to noisy/crowded embedding spaces.

Large-Batch Effect (CLIP-Scale Training)

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CLIP Original Training

- Batch size $N = 32,768$
- Each image/text paired with 32k negatives.
- Requires massive compute (thousands of GPUs).

Example: $N = 4$ (Small Scale)

$$S = \begin{bmatrix} 0.98 & 0.14 & 0.2 & 0.1 \\ 0.98 & -0.14 & 0.3 & 0.4 \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

- Hard negatives (e.g., $S_{1,3} = 0.2$) dominate learning.

Why Large Batches Help

- More negatives $\rightarrow$ better estimate of true distribution.
- Exposes model to diverse failure modes.

Zero-Shot Classification with CLIP

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Given candidate classes $\{t_k\}_{k=1}^K$:

$$p(y = k|x) = \frac{\exp(s(f_I(x), f_T(t_k))/\tau)}{\sum_j \exp(s(f_I(x), f_T(t_j))/\tau)}$$

Interpretation

Text embeddings act as classifier weights.

Prompting Trick

Instead of label "dog", use:

"a photo of a dog"

Prompt ensembling improves performance.

Hands-on Coding with the Inference CLIP models

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CLIP and CLIPSeg

- Image + text (proposed classes) + CLIP model = one-shot classification. The code is available via the [link #1](#).
- Image + text (proposed classes) + CLIPSeg model = one-shot semantic segmentation. The code is available via the [link #2](#).

CLIP is Followed by...

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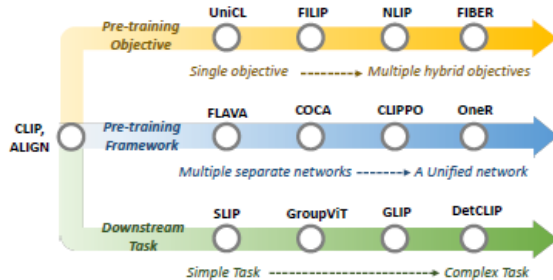


Figure: Illustration of development of VLMs for visual recognition [Zhang et al., 2024]

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Section 5. VLM's Objectives (losses)

VLM Pretraining Objectives

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Three Categories

- **Contrastive**: Align visual and textual embeddings via pairwise similarity
- **Generative**: Predict masked content or generate text given vision
- **Alignment**: Match global (image-text) or local (region-word) pairs

Setup

Let $\mathcal{D} = \{x_n^I, x_n^T\}_{n=1}^N$, $z^I = f_\theta(x^I)$, $z^T = f_\phi(x^T)$.

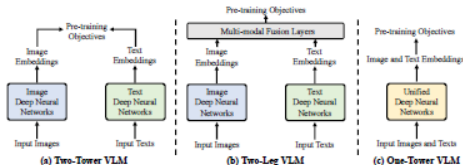


Figure: Typical VLM frameworks [Zhang et al., 2024]

Contrastive Objectives (1)

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Image Contrastive Learning (InfoNCE)

$$\mathcal{L}_{\text{InfoNCE}}^I = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(z_i^I \cdot z_+^I / \tau)}{\sum_{j \neq i} \exp(z_i^I \cdot z_j^I / \tau)}$$

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$$\mathcal{L}_{\text{InfoNCE}}^{I \rightarrow T} = -\frac{1}{B} \sum_i \log \frac{\exp(z_i^I \cdot z_i^T / \tau)}{\sum_j \exp(z_i^I \cdot z_j^T / \tau)}$$

$$\mathcal{L}_{\text{InfoNCE}}^{T \rightarrow I} = -\frac{1}{B} \sum_i \log \frac{\exp(z_i^T \cdot z_i^I / \tau)}{\sum_j \exp(z_i^T \cdot z_j^I / \tau)}$$

Contrastive Objectives (2)

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Image-Text-Label Contrastive Learning (UniCL) [Khosla et al., 2021]

$$\mathcal{L}_{\text{label}}^{I \rightarrow T} = - \sum_i \frac{1}{|P(i)|} \sum_{k \in P(i)} \log \frac{\exp(z_i^I \cdot z_k^T / \tau)}{\sum_j \exp(z_i^I \cdot z_j^T / \tau)}$$

where τ is a temperature hyperparameter, $P(i)$ is the set of positive labels for sample i

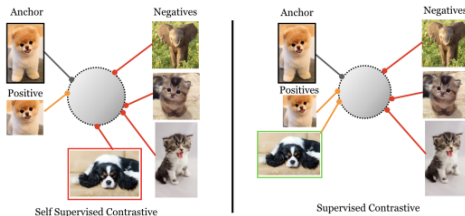


Figure: Supervised vs. self-supervised contrastive losses [Khosla et al., 2021]

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Generative Objectives (1)

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Masked Image Modeling (MIM)

$$\mathcal{L}_{\text{MIM}} = -\frac{1}{B} \sum_i \log f_{\theta}(x_i^I | \hat{x}_i^I)$$

Explanation

- Input image x_i^I is split into patches, some patches are masked: $\hat{x}_i^I$
- The model reconstructs the masked parts with the log-likelihood loss of correct reconstruction



Figure: Masked Image Modeling (MIM) intuition [Khosla et al., 2021]

Generative Objectives (2)

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Masked Language Modeling (MLM)

$$\mathcal{L}_{\text{MLM}} = -\frac{1}{B} \sum_i \log f_{\phi}(x_i^T | \hat{x}_i^T)$$

Masked Cross-Modal Modeling (MCM)

$$\mathcal{L}_{\text{MCM}} = -\frac{1}{B} \sum_i \left[\log f_{\theta}(x_i^I | \hat{x}_i^I, \hat{x}_i^T) + \log f_{\phi}(x_i^T | \hat{x}_i^I, \hat{x}_i^T) \right]$$

Image-to-Text Generation

$$\mathcal{L}_{\text{ITG}} = -\sum_{l=1}^L \log f_{\theta}(x_l^T | x_{<l}^T, z^l)$$

Alignment Objectives

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Image-Text Matching (ITM)

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$$\mathcal{L}_{\text{ITM}} = p \cdot \log S(z^I, z^T) + (1 - p) \cdot \log(1 - S(z^I, z^T))$$

where $p \in 0, 1$ is label indicating match (1) or mismatch (0).

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Region-Word Matching (RWM)

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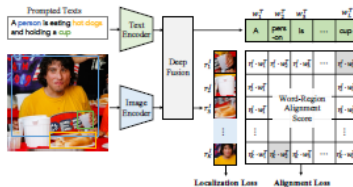
$$\mathcal{L}_{\text{RW}} = p \cdot \log S_r(r^I, w^T) + (1 - p) \cdot \log(1 - S_r(r^I, w^T))$$

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Section 6. Fine-tuning of Large Models

Strategies for LMs

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Comparison of Parameter-Efficient Methods

Method	Added Params	Modifies Forward Pass	Key Advantage
Full FT	100%	$h = (W_0 + \Delta W)x$	Highest accuracy
LoRA	$\sim 0.1\%$	$h = W_0x + BAx$	Balance of efficiency/performance
Adapter	$\sim 1\%$	$h = W_0x + W_2(\sigma(W_1x))$	Modular
Prefix Tuning	$\sim 0.5\%$	$[P; x] \rightarrow \text{Attention}$	No backbone changes
BitFit	$\sim 0.01\%$	$h = W_0x + b$	Only biases updated

Mathematical Forms

- **Adapter:** $W_2 \in \mathbb{R}^{d \times r}$, $W_1 \in \mathbb{R}^{r \times d}$
- **Prefix Tuning:** $P \in \mathbb{R}^{l \times d}$ (prepended tokens)
- **BitFit:** $b \in \mathbb{R}^d$ (bias terms only)

When to Use LoRA?

- Need high parameter efficiency ($r \leq 64$)
- Preserve original model architecture
- Balance between compute and accuracy

LoRA: Low-Rank Adaptation [Hu et al., 2021]

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Key Mathematical Formulation

For a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$:

$$W = W_0 + \underbrace{BA}_{\text{Low-rank update}} \quad \begin{cases} B \in \mathbb{R}^{d \times r} \\ A \in \mathbb{R}^{r \times k} \\ r \ll \min(d, k) \end{cases}$$

Example: 1024x1024 Layer with Rank=8

- Original params: $1024 \times 1024 = 1,048,576$
- LoRA params: $8 \times (1024 + 1024) = 16,384$
- Reduction: $\frac{16,384}{1,048,576} \approx 1.56\%$

LoRA in Action

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Forward Pass Computation

For input $x \in \mathbb{R}^k$:

$$h = W_0 x + \underbrace{B(Ax)}_{\text{Rank-constrained update}}$$

Gradient Flow

- Frozen weights: $\nabla_{W_0} \mathcal{L} = 0$
- Adaptor gradients:

$$\nabla_B \mathcal{L} = (\nabla_h \mathcal{L}) x^\top A^\top \quad \nabla_A \mathcal{L} = B^\top (\nabla_h \mathcal{L}) x^\top$$

Why This Works

- Preserves pretrained knowledge (W_0 fixed)
- Efficient training (only update B, A)
- Low-rank bottleneck prevents overfitting

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Paper reading

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Vision-Language Models for Vision Tasks: A Survey

Most visual recognition studies rely heavily on crowd-labelled data in deep neural networks (DNNs) training, and they usually train a DNN for each single visual recognition task, leading to a laborious and time-consuming visual recognition paradigm. To address the two challenges, Vision-Language Models (VLMs) have been intensively investigated recently, which learns rich vision-language correlation from web-scale image-text pairs that are almost infinitely available on the Internet and enables zero-shot predictions on various visual recognition tasks with a single VLM. This paper provides a systematic review of visual language models for various visual recognition tasks, including: (1) the background that introduces the development of visual recognition paradigms; (2) the foundations of VLM that summarize the widely-adopted network architectures, pre-training objectives, and downstream tasks; (3) the widely-adopted datasets in VLM pre-training and evaluations; (4) the review and categorization of existing VLM pre-training methods, VLM transfer learning methods, and VLM knowledge distillation methods; (5) the benchmarking, analysis and discussion of the reviewed methods; (6) several research challenges and potential research directions that could be pursued in the future VLM studies for visual recognition. [Zhang et al., 2024].

Future Directions for VLMs [Zhang et al., 2024]

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Challenges in Pretraining

- **Fine-grained vision-language correlation:** needed for zero-shot dense prediction
- **Unified vision-language modeling:** one transformer for both image and text
- **Multilingual pretraining:** remove language/cultural bias
- **Data-efficient learning:** less data, more supervision between pairs
- **Leveraging LLMs:** augment image-text pairs using rich linguistic knowledge

Challenges in Transfer Learning

- **Unsupervised transfer:** reduce overfitting to few-shot labels
- **Visual prompts/adapters:** complement text prompting
- **Test-time adaptation:** update prompts dynamically at inference
- **Transfer with LLMs:** generate task-specific prompts automatically

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